Title: Integrating time of day dynamics of gene expression with metabolic networks to improve crop performance

Abstract: We are facing a projected need of doubling our current food supply by 2050[1,2]. This incredible challenge is compounded by the effects of climate change that will expose crops to novel environmental stressors. To meet this challenge, we need crops that produce more yield while tolerating variable stress conditions. Many plant responses to stress are driven by genes that operate on a ~24-hour cycle. My research aims to determine how time of day influences plant stress response, with the ultimate goal of improving crop performance in a changing climate.

Background: Understanding how crops respond to climate change is critical to our ability to meet growing food supply demands. Modifying gene expression underlying physiological (physical functions that allow an organism to survive) and/or metabolic (sum of all chemical reactions that convert food into energy) processes is one way that organisms respond to stress[3]. Many biological processes are regulated by an internal timekeeper known as the circadian clock, which adjusts the timing of peak gene expression to specific times of day to produce coordinated biological responses[4]. For example, many plants open stomata (pores on the leaf for gas exchange) during the day and close them at night. This allows plants to take up CO_2 and photosynthesize while the sun is up and make starch for energy at night. However, stomata also release water while open which can be detrimental under drought. Proper stomatal regulation requires tight control of gene expression to balance the need to photosynthesize while simultaneously avoiding water loss[5,6]. This is critical at midday under drought conditions when the sun and temperatures are highest. However, few studies integrate time of day into their design, failing to capture dynamic changes in gene expression and physiology. To fully understand how plants optimize stress responses, we must capture not only *what* they are changing, but *when* they are changing it.

The clock also coordinates metabolism with a plant's environment by regulating nutrient pools to ensure availability when they are needed most[7]. Under stress, plant nutrients are redirected to various biological pathways to mount a defense, thereby altering nutrient availability. A major goal in plant science is to improve how plants regulate their nutrient/metabolite pools, particularly during stress (e.g. drought; frost). Metabolic networks, or interconnected 'pathways' of biochemical reactions (e.g. Citric Acid Cycle), provide a snapshot of metabolism that allows us to track where nutrients are being directed within the network at a given point in time. This is especially important for limited nutrients such as sulfur, an integral component in plant health and defensive compounds[8] including glucosinolates. Glucosinolates (GSLs) are sulfur and nitrogen rich metabolites found in the mustard family (*Brassicaceae*) and are responsible for their characteristic flavor (mustard, wasabi, brussel sprouts)[9]. Previous research shows that GSL content oscillates throughout the day[10], potentially acting as a source of sulfur[11] which can be reallocated towards stress responses and/or growth. How this turnover is regulated is unknown. Sulfur deficiencies in crops have increased worldwide due to a lack of sulfur in the soil[2,12], exacerbating the need to improve sulfur-use efficiency in our crops. This is vital for *Brassicaceae* crops like Canola that have high sulfur demands due to production of GSLs. <u>My thesis research addresses these challenges by integrating temporal dynamics of gene expression and metabolite abundance to build more predictive models for targeted breeding efforts to improve stress responses in crops</u>.

1. Elucidate time of day based regulation of gene expression underlying differential growth responses to water limitation in *Brassica napus* (Canola)

Drought is one of the most detrimental environmental stressors for crops[13]. While most drought studies focus on how plants respond at the onset of water limitation, few examine how plants adapt to prolonged water deficit which may be more aligned with the effects of climate change. I sought to elucidate temporal changes to physiology and metabolism in Canola after prolonged exposure to water limitation and to identify what genes were important to this process. Overall, I found that while cultivars tended to modify genes involved in similar processes (carbon metabolism; photosynthesis), the time of day that expression was altered differed. For example, two cultivars with differential tolerance to water limitation altered expression of genes involved in GSL biosynthesis but the timing of the change differed by at least six hours (Fig. 1: pink panels). This indicated significant differences in regulation strategies of sulfur metabolism. To associate changes in gene expression with the metabolites, I captured temporal differences in GSL content using a novel collection, extraction, and mass spectrometry workflow I developed. This confirmed that the more tolerant line had higher levels of GSLs under prolonged water limitation, highlighting the importance of targeting aspects of sulfur metabolism to engineer more drought tolerant crops. Pilot data collected from these experiments were used to secure a Department Of Energy grant, described in chapter 3. Through this work I have also discovered evidence for differential GSL regulation on a spatial scale (Fig. 2), which challenges our current understanding of GSLs as most studies collect a small portion of the leaf and are likely to be missing key spatial differences in metabolite content.

To pursue this further I wrote and was awarded an MnDRIVE fellowship that has been supplemented by a National Science Foundation grant. I am first author on this chapter which is in the submission phase. I have presented these findings at multiple conferences, including an invited talk at a national conference (American Society of Plant Biology; 2022).

2. Identify gene regulatory elements underlying differential regulation of cold tolerance in *Brassica rapa* (e.g. Chinese cabbage, Pak choi, oilseeds)

Late cold snaps can severely damage crops[14], particularly if they happen early in the growing season when plants are young. Identifying genes underlying cold/freezing tolerance is a major goal in plant breeding. Most crops, including *B. rapa,* have undergone whole genome duplications resulting in multiple gene copies. This complexity has challenged our ability to identify breeding targets and to understand how these expanded genomes regulate stress responses through alterations in gene expression.

In this chapter I seek to identify which regulatory genes are important for cold stress tolerance and what regulatory elements (sequences upstream of the gene that control activity) are shared among cold tolerant *B. rapa* cultivars. I have adapted a time-of-day based RNA-sequencing pipeline (a set of computational scripts that process, analyze, and visualize data) to identify which gene copy/copies are cold responsive and what time of day they are being altered. I have found that while crop types (Chinese cabbage, Pak choi, oilseeds) share some of the same cold responsive genes, the copy they use often differs among individuals. This is also the case *within* a crop type, suggesting divergence in regulatory elements within *B. rapa*. Identifying these regulatory elements provides the opportunity to genetically modify a plant's cold tolerance across variable crop types. In collaboration with an undergraduate who I mentored, we have generated predictive models to identify candidate cold tolerant regulatory genes and are using them to identify shared regulatory elements.

This research is part of a larger project that is funded by a Joint Genomes Institute grant (CSP-504419) and a National Science Foundation grant (IOS-2029549). I presented initial findings at a Cold Spring Harbor Laboratory conference (Fall 2023) and was invited to present this research at a lab at Stanford University (Spring 2024). I plan to submit these findings for publication as first author this summer (2024).

3. Build temporally informed metabolic networks to reconstruct the glucosinolate pathway across the *Brassicaceae* (Mustard family)

As we found in chapter 1, regulation of GSLs is contributing to stress responses but the exact role is unclear. To improve sulfur use efficiency in crops we need to reconstruct (model) the GSL/sulfur pathway in crop species to identify which enzymes to target. Although we have some understanding of how model plants (Arabidopsis; plant version of *E. coli*) regulate sulfur we are unclear on what enzymes participate in GSL turnover, how this affects total sulfur pools, or if there are differences in the GSL pathway/sulfur regulation among species. In the final chapter of my thesis, I aim to bridge these gaps in knowledge by pairing temporally informed (28 hour) gene expression with metabolite levels in three *Brassicaceae* species.

This chapter is in the data processing and analysis phase. Final steps include running previously collected time series data through the computational pipelines I described above to correlate time-of-day specific changes in gene expression with corresponding changes to the metabolites these genes control. From previous experiments I found that only a subset of GSLs oscillate in *B. rapa*. Identifying the genes that control oscillating GSLs will be imperative to our ability to engineer specific GSL levels in a variety of economically important crops. Overall, my research deepens our understanding of how genetically complex plants optimize stress responses, but we need more robust metabolic models to understand how these dynamic responses alter nutrient pools. To accomplish this goal I need more time to focus my efforts on interpreting this complex pathway in the two remaining *Brassicaceae* species. Receiving this fellowship will allow me to achieve this goal as well as others described below.

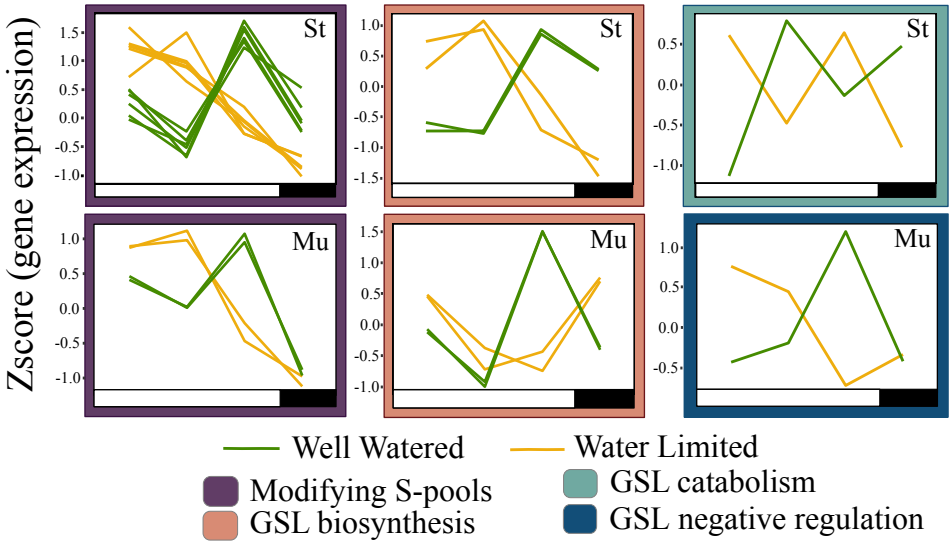
Future Goals and Impact of Doctoral Dissertation Fellowship (DDF)

My path to where I am today has been anything but traditional. I did not grow up dreaming about becoming a scientist; while I enjoyed science and math my goal was to work in theater. It wasn't until I began taking classes at a small community college that I became inspired to pursue a scientific career. I wouldn't be where I am today if not for two faculty members that recognized my passion and dedication to learning. Their belief in me showed me the importance of role models, and I'm passionate about inspiring future generations. I have pursued this passion by closely advising five undergraduate students and served on the board of our graduate student group (Phytograds) for two years. I believe that building a diverse community is critical to our success as scientists and have initiated a number of department wide team building events, as well as a number of outreach activities to engage the local community with science.

Upon completion of my degree I will be first author on at least three manuscripts and co-author on two others. I hope to continue on in the circadian clock field as a postdoctoral researcher in a computationally intensive lab. However, I have yet to have the time to dedicate to networking and/or showcasing my more recent research. Being awarded the DDF will be critical for achieving this goal and will allow me the time and resources to focus on interpreting results and writing up my final chapter. Overall, this fellowship will facilitate my abilities as a scientist to continue to create new tools that help us better understand the dynamic world around us.

Appendix

Figure 1. B. napus cultivars alter sulfur metabolism and glucosinolate (GSL) related gene expression at different times of day in response to prolonged water limitation. Expression patterns of genes enriched in modifying sulfur pools (purple), GSL biosynthesis (pink), and reduction of GSLs (aqua; blue) as indicated by colored boxes. Each line is a single enriched gene. Bars indicate day (white) and night(black).



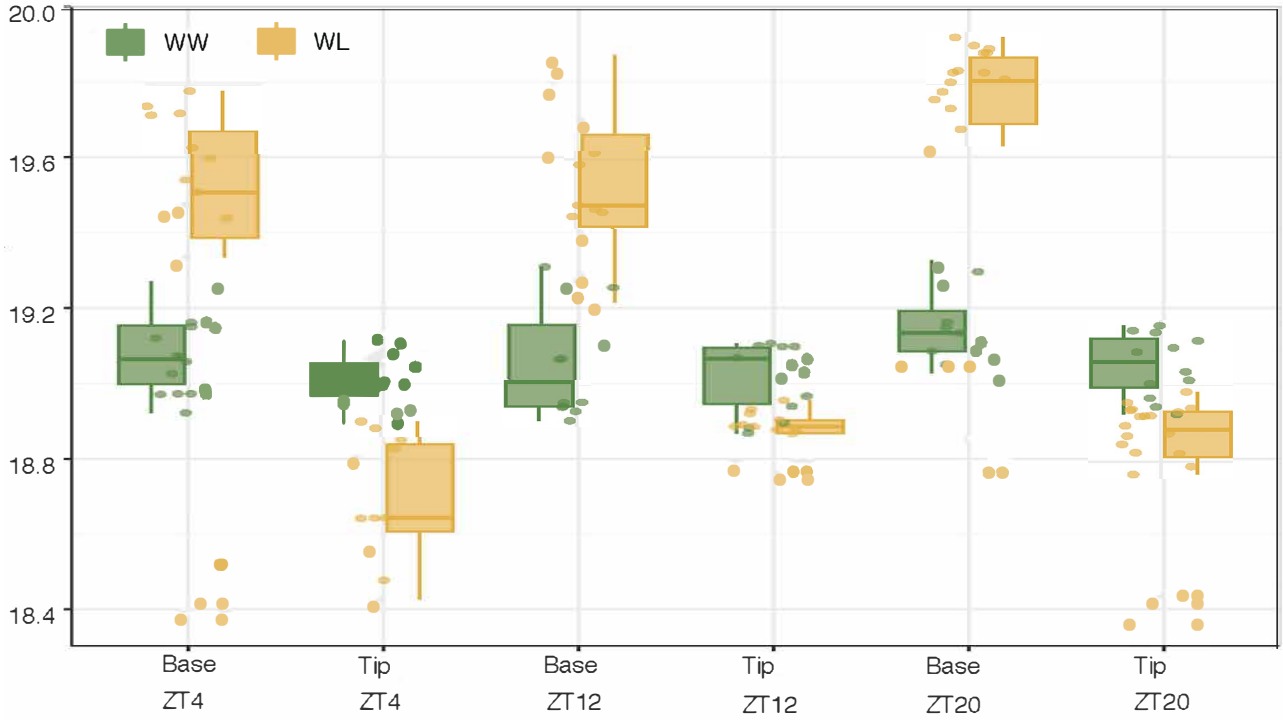


Figure 2. Spatial differences in glucosinolate content (gluconapin). The amount of gluconapin does not vary under control conditions (WW) but does after prolonged water limitation (WL). However, the direction of the change (higher/lower) depends on where on the leaf it was collected (base or tip). ZT; hours after lights on. Each point is an individual plant.

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